

MERCED, CA, USA

WMO: 724815

Lat: 37.285N

Lon: 120.513W

Elev: 46

StdP: 100.77

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 23257

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.3	0.1	-5.8	2.3	9.3	-3.6	2.8	8.1	11.0	12.1	9.2	11.2	1.1	100	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	38.6	20.9	37.2	20.4	35.9	19.8	22.3	35.5	21.4	34.5	20.6	33.4	4.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.6	27.6	16.4	11.7	26.6	15.5	11.1	26.2	65.5	35.6	62.1	34.4	59.3	33.5	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.5	-3.3	41.2	1.3	1.1	-4.3	42.0	-5.1	42.6	-5.8	43.3	-6.8	44.1		
			-4.1	24.1	1.5	2.3	-5.2	25.8	-6.1	27.1	-7.0	28.4	-8.1	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	8.0	10.1	12.9	15.4	19.7	23.8	26.4	25.4	23.0	17.8	11.9	8.0
	DBStd	7.14	2.69	2.60	3.05	3.53	3.45	3.33	2.75	2.38	3.26	3.11	3.08	2.88
	HDD10.0	203	73	28	9	2	0	0	0	0	0	0	16	75
	HDD18.3	1407	319	229	169	101	24	2	0	0	4	47	192	319
	CDD10.0	2726	12	32	98	165	302	413	508	477	390	242	74	14
	CDD18.3	889	0	0	1	13	68	165	249	218	144	31	1	0
	CDH23.3	12512	0	0	53	282	1067	2342	3459	2905	1901	485	18	0
	CDH26.7	6769	0	0	7	91	467	1289	2077	1676	1003	157	2	0
Wind	WSAvg	2.7	2.1	2.4	2.6	3.1	3.5	3.6	3.4	3.1	2.6	2.3	1.9	2.2
Precipitation	PrecAvg	315	61	57	51	25	11	2	1	1	4	18	41	48
	PrecMax	550	180	155	139	117	54	16	11	12	34	96	125	114
	PrecMin	167	4	4	0	0	0	0	0	0	0	0	0	2
	PrecStd	100	45	40	38	24	14	4	2	2	8	22	32	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	21.9	27.5	32.9	36.4	39.9	40.7	39.6	37.9	33.8	25.7	18.3
		MCWB	11.8	14.3	16.3	18.0	19.5	21.4	21.7	20.8	19.8	18.4	15.7	12.8
	2%	DB	16.6	19.7	24.7	29.1	34.0	37.5	38.8	37.6	36.5	30.5	22.8	16.3
		MCWB	11.6	13.5	15.4	16.6	18.6	20.8	20.9	20.5	19.3	17.5	14.5	11.7
	5%	DB	14.8	17.8	22.6	26.7	31.7	35.7	37.3	36.2	34.6	28.0	20.7	14.7
		MCWB	11.1	12.5	14.7	15.8	17.5	19.7	20.8	19.9	18.8	16.3	13.9	11.2
	10%	DB	13.0	16.1	20.4	23.9	29.0	33.5	35.7	34.5	32.6	26.1	18.4	13.3
		MCWB	10.3	11.7	13.8	14.7	16.5	18.9	20.1	19.4	18.4	15.5	12.8	10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	15.7	18.1	19.5	21.3	23.0	23.7	22.9	22.4	19.9	16.9	14.9
		MCDB	16.3	19.7	24.5	28.3	34.0	36.8	36.3	35.6	32.8	29.0	23.2	17.2
	2%	WB	12.8	14.4	16.5	17.7	19.5	21.7	22.4	21.7	20.7	18.6	15.5	13.2
		MCDB	14.8	18.2	23.1	27.4	32.1	35.2	36.3	34.8	32.5	27.9	21.3	15.0
	5%	WB	11.7	13.2	15.3	16.4	18.2	20.6	21.5	20.7	19.8	17.4	14.4	12.1
		MCDB	13.5	16.9	21.2	25.5	29.8	33.4	35.0	33.7	32.4	26.7	19.2	13.8
	10%	WB	10.7	12.2	14.1	15.3	17.1	19.7	20.6	20.0	18.9	16.3	13.4	10.9
		MCDB	12.9	15.4	19.4	22.9	27.9	32.1	33.8	32.8	31.2	24.2	17.6	12.7
Mean Daily Temperature Range	5% DB	MDBR	9.4	11.4	13.0	14.3	16.0	17.4	17.9	18.0	17.5	15.3	12.5	9.9
		MCDBR	12.7	14.2	16.7	18.7	19.6	19.5	19.0	19.5	20.0	18.5	15.9	12.0
		MCWBR	8.3	8.6	8.6	8.1	7.6	6.7	6.4	6.5	6.7	7.9	8.4	7.7
	5% WB	MCDBR	9.2	12.4	15.3	17.3	18.4	18.5	17.8	18.4	18.4	16.2	13.5	8.1
		MCWBR	6.5	7.9	8.5	7.9	7.6	6.8	6.5	6.6	6.7	7.0	7.6	5.7
Clear-Sky Solar Irradiance	taub	0.347	0.352	0.359	0.361	0.361	0.352	0.366	0.355	0.356	0.367	0.355	0.335	
	taud	2.384	2.393	2.369	2.381	2.402	2.436	2.409	2.417	2.425	2.378	2.373	2.445	
	Ebn at Noon	828	876	903	919	920	924	908	914	895	846	811	820	
	Edh at Noon	90	101	113	118	118	114	116	113	106	101	90	80	
All-Sky Solar Radiation	RadAvg	2.35	3.45	4.93	6.31	7.55	8.40	8.17	7.41	6.10	4.43	2.95	2.15	
	RadStd	0.38	0.38	0.44	0.50	0.38	0.27	0.25	0.21	0.19	0.25	0.18	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (45 neighbors)	N/A	N/A	-1.15	N/A	N/A	N/A	N/A	-114	+138	+35

Nomenclature: See separate page